

Tridib Banik

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Highlights of Qualifications

- Software Engineering Co-op undergrad at McMaster; available for a 4-month internship starting in Spring '26.
- Active team member in the Battery Workforce Challenge (BWC) Software sub-team for 20+ months, focusing on model-based design, system integration, and simulation of a full-scale battery pack for a light-duty EV.

Education

Bachelor of Engineering, Software Engineering Co-op
McMaster University, Hamilton, ON

Sept. 2023 – Apr. 2028
CGPA: 11.28 / 12.0

- **Relevant Courses:** Dynamic Systems and Control (MATLAB, Simulink), Signals and Systems, Software Requirements & Security, Databases (SQL), Operating Systems (C), OOP (Java), Large System Design (UML).

Skills

Languages: MATLAB, Java, C, Python, SQL (IBM DB2), HTML, CSS, Bash, ARM64, Verilog.

Tools: GitHub, GitLab CI/CD, JUnit, Maven, Linux, UML, $\text{\LaTeX}$, Simulink, Excel, PowerPoint.

Methodologies: Agile (scrum meetings, sprint deliveries), Software Dev Life Cycle, Requirements Tracing.

Work Experience

BWC Software Team Member and Lab Assistant, CMHT Lab

Sept. 2024 – Present

- Tested 100% of assigned subsystem requirements, passing 95%+ of unit test cases; contributed to software/hardware documentation, release notes, system diagrams, and MATLAB signal behaviour figures; attended 35+ weekly team lead meetings over 9 months to discuss risks and mitigation strategies.
- Authored a 308-page internal PowerPoint guide detailing software installation and hardware setup for a custom BMS (CMU, BMU, BJB, Pack Emulator) to support onboarding and research continuity.
- Configured a 5-stage GitLab CI pipeline for Simulink model verification, registering a GitLab Runner and managing CI/CD variables; authored a guide covering setup, directory structure, and MATLAB commands.

Research Assistant, CMHT Lab, McMaster Automotive Resource Centre

June 2025 – Aug. 2025

- Co-authored research (under review) analyzing memory usage of a safety-critical EV BMS on an automotive-grade microprocessor, benchmarking 7 major functions, revealing BMS functions use only 0.99% flash and 9.23% RAM, with State of Charge alone accounting for 54.03% and 56.72%, respectively.
- Designed, validated and documented 4 model-based control logic modules (Contactor Control, Cell Balancing, AC/DC Fast Charging, Fault Diagnosis) using Simulink, ensuring compliance with software requirements.

Teaching Assistant (1P13: Integrated Cornerstone Design Projects in Engineering)

Jan. 2026 – Apr. 2026

- Mentored 23 students across 5 teams through a semester-long material selection project spanning 6 milestones and a final research presentation, delivering live verbal feedback to every team and individual on objective trees, metrics, regulatory compliance, and decision matrices for wastewater filtration systems.

Projects

Smart City Environmental Monitoring and Alert System (SCEMAS)

Jan. 2026 – Apr. 2026

- Designed and justified a heterogeneous architecture (PAC overall; Pipe-and-Filter, Blackboard, and Repository across 3 subsystems) for a smart city environmental monitoring platform, producing Analysis Class Diagrams, CRC cards, and a high-level system architecture diagram across 4 structured deliverables.

TrafficLightRL | Python, OpenAI Gymnasium, Stable-Baselines3 in PyTorch

Jan. 2025 – Mar. 2025

- Architected a modular reinforcement learning system with separated environment, training, and evaluation components; results averaged over 100 episodes per data point, demonstrating up to 29.55% wait time reduction and 35.56% emissions reduction at high traffic density compared to a traditional control system.
- Implemented logging, metrics tracking, and automated experiment evaluation across 3 traffic densities, presenting results at the Canadian Undergraduate Conference on AI (320+ attendees).